Distributed Systems
Section 3. Architectures

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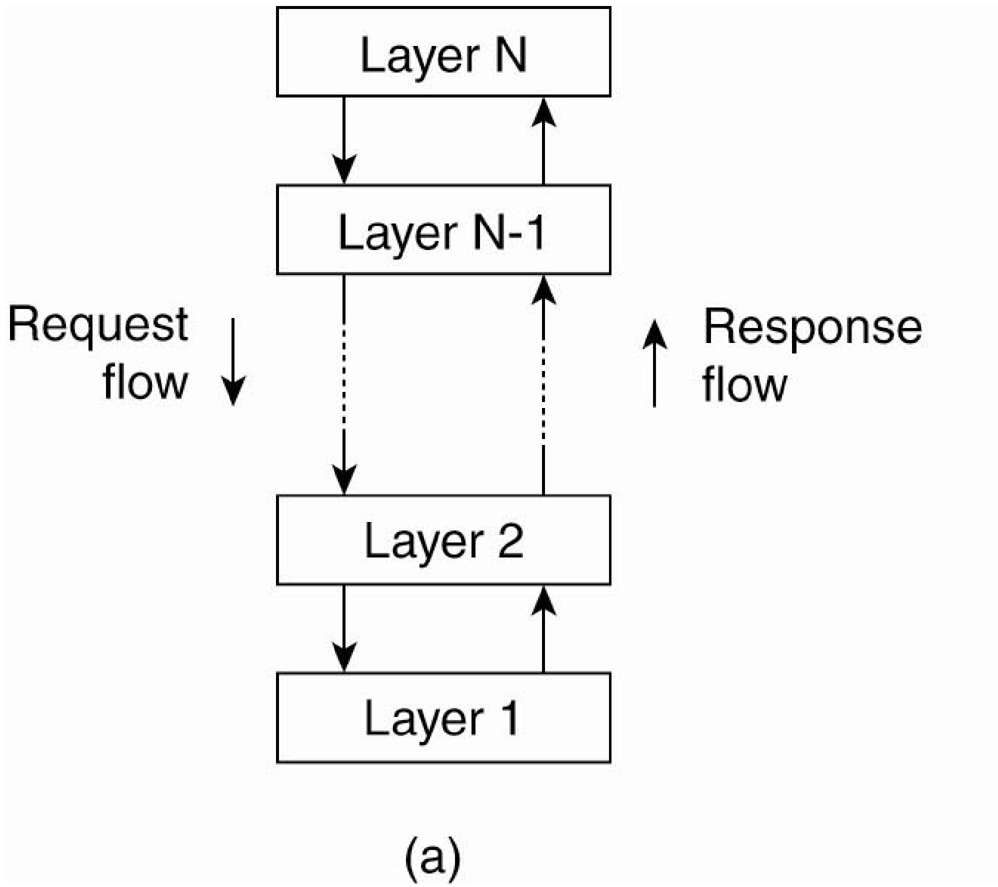
Generations of Distributed Systems

Distributed systems:	*Early*	*Internet-scale*	*Contemporary*
Scale	Small	Large	Ultra-large
Heterogeneity	Limited (typically relatively homogenous configurations)	Significant in terms of platforms, languages and middleware	Added dimensions introduced including radically different styles of architecture
Openness	Not a priority	Significant priority with range of standards introduced	Major research challenge with existing standards not yet able to embrace complex systems
Quality of service	In its infancy	Significant priority with range of services introduced	Major research challenge with existing services not yet able to embrace complex systems

Architectural Styles (1)

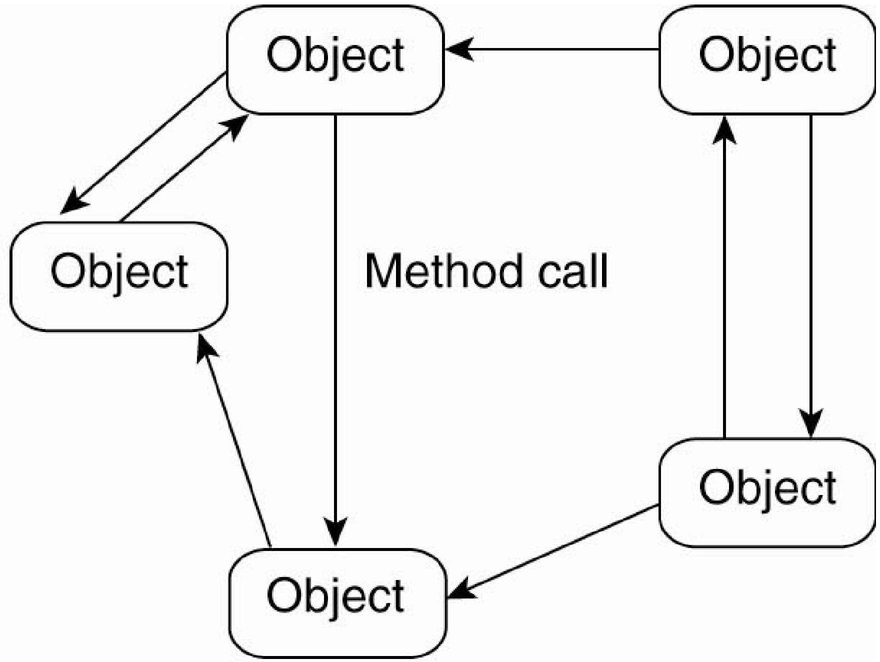
- Important styles of architecture for distributed systems
- Layered architectures
- Object-based architectures
- Data-centered architectures
- Event-based architectures

Architectural Styles (2)



- The (a) layered architectural style

Architectural Styles (3)

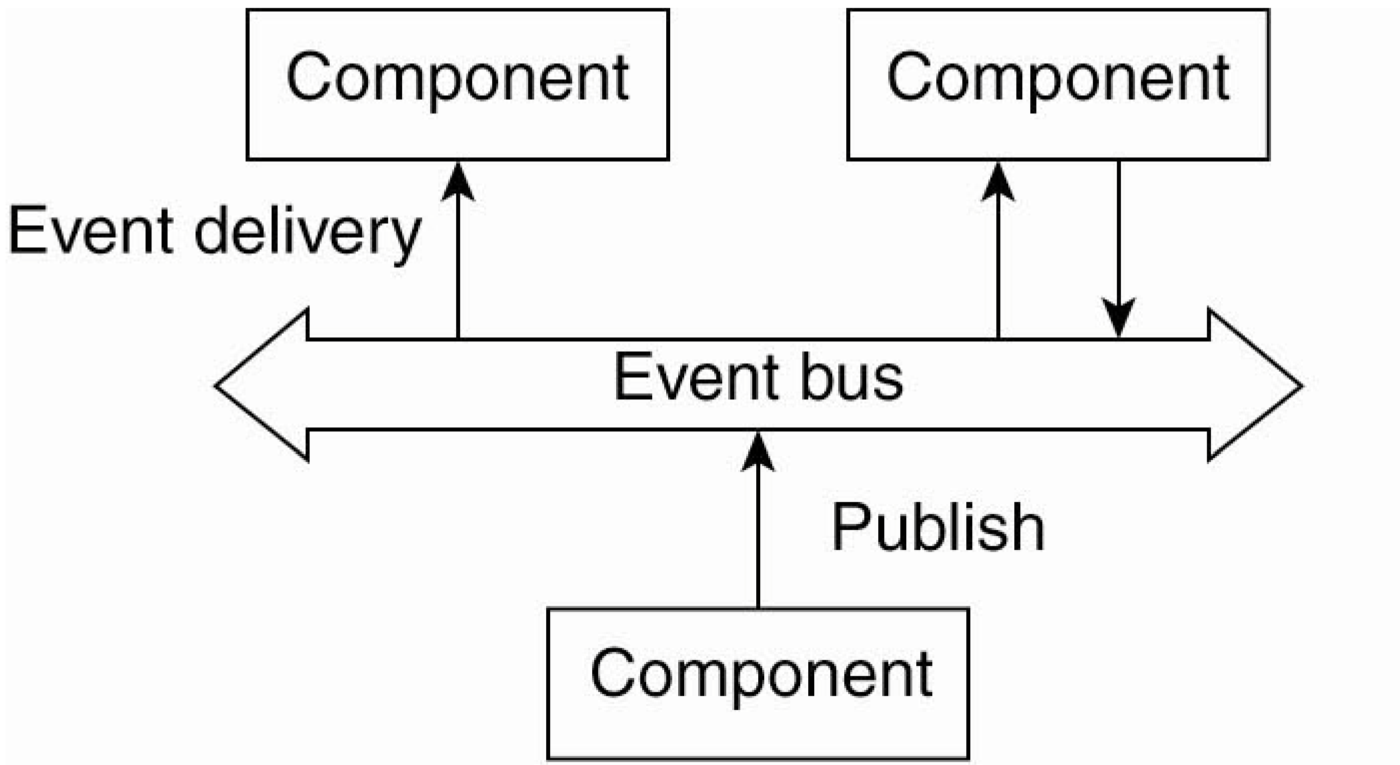


(b)

- The object-based architectural style.

Architectural Styles (4)

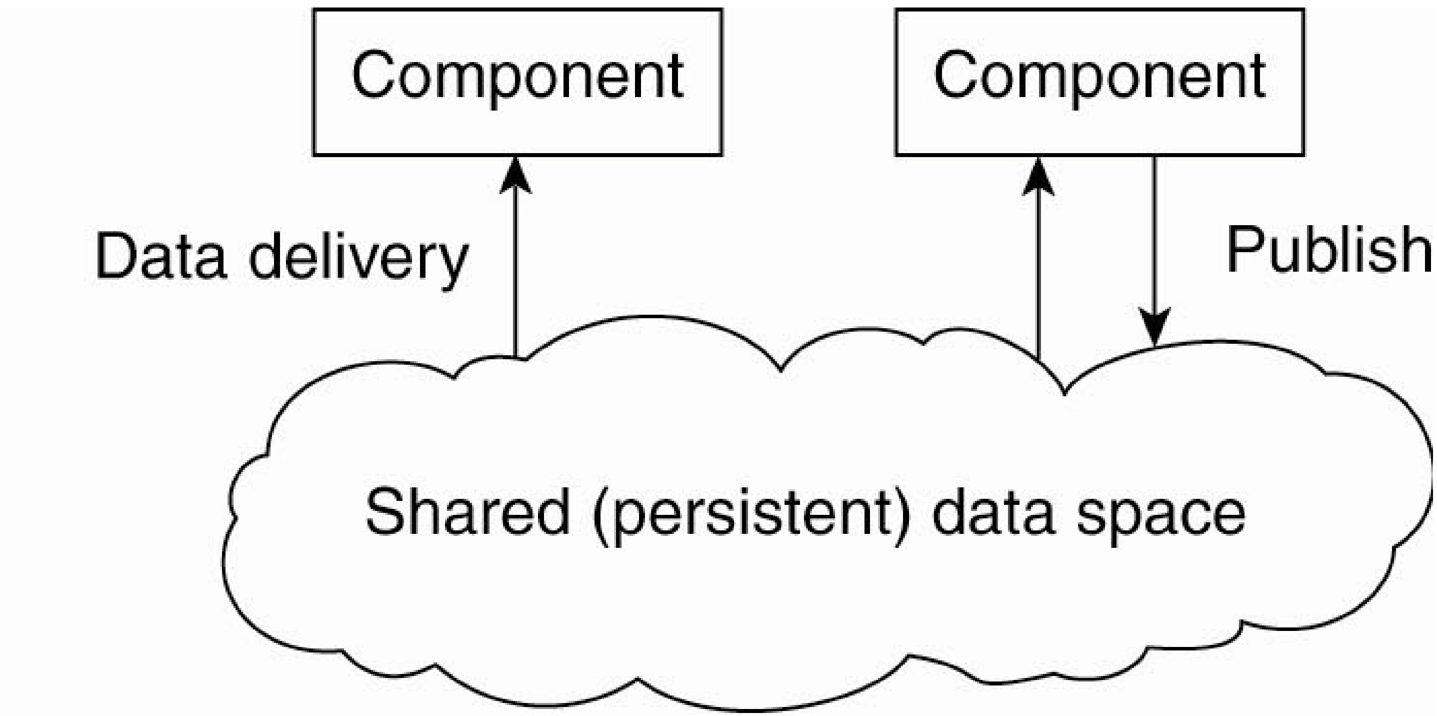
- The event-based architectural style and …



(a)

Architectural Styles (5)

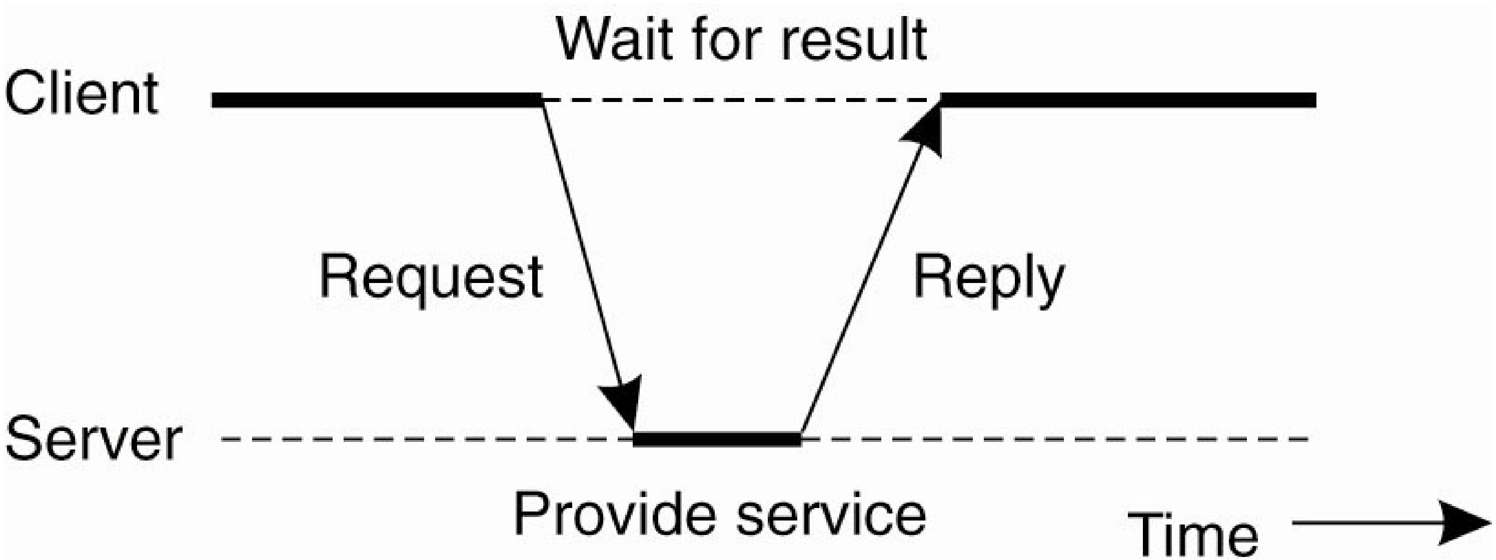
- The shared data-space architectural style.



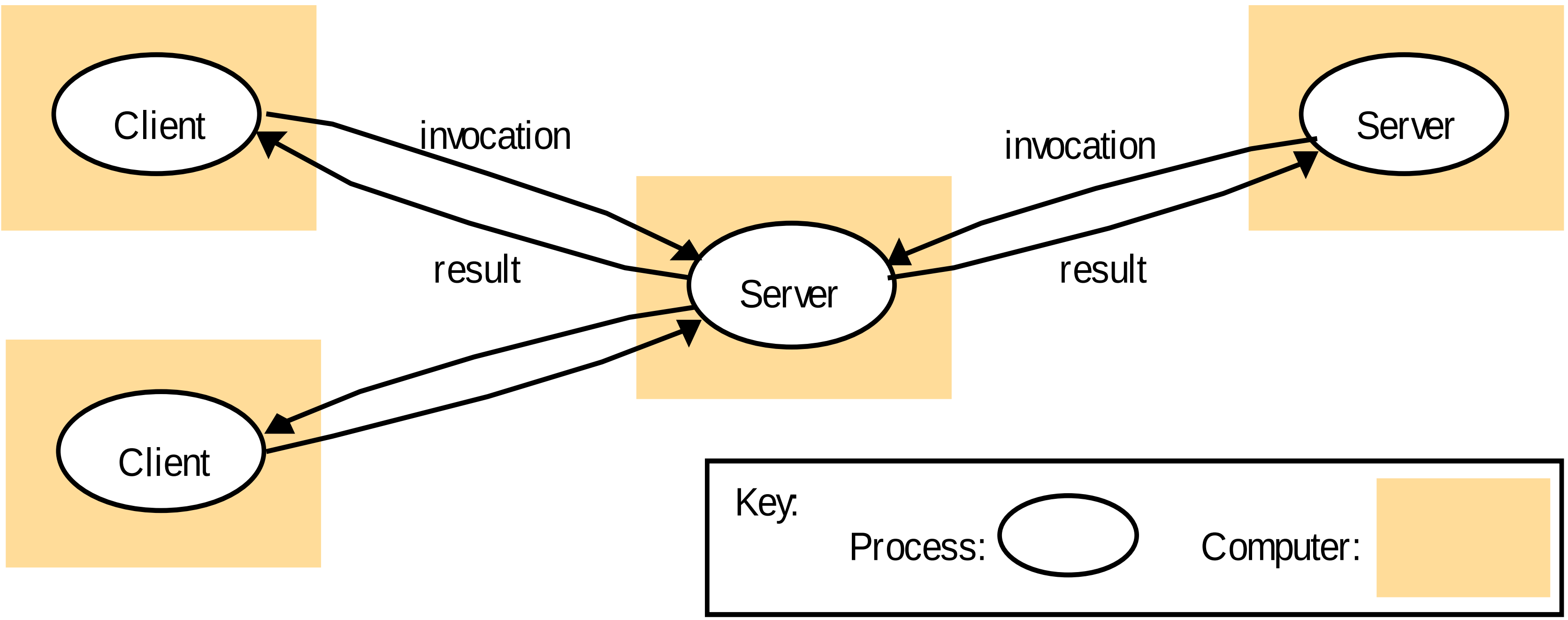
(b)

Centralized Architectures

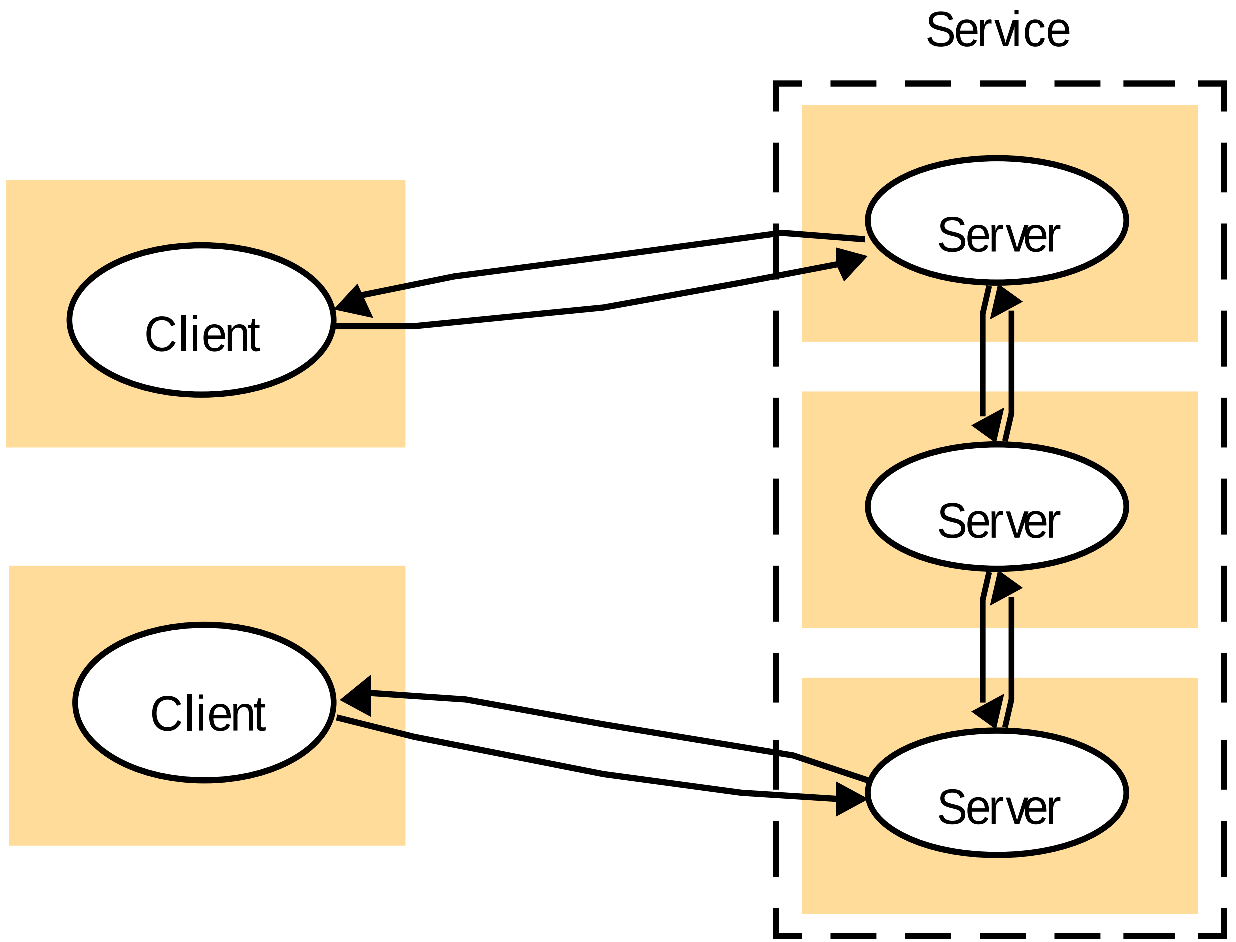
- General interaction between a client and a server.



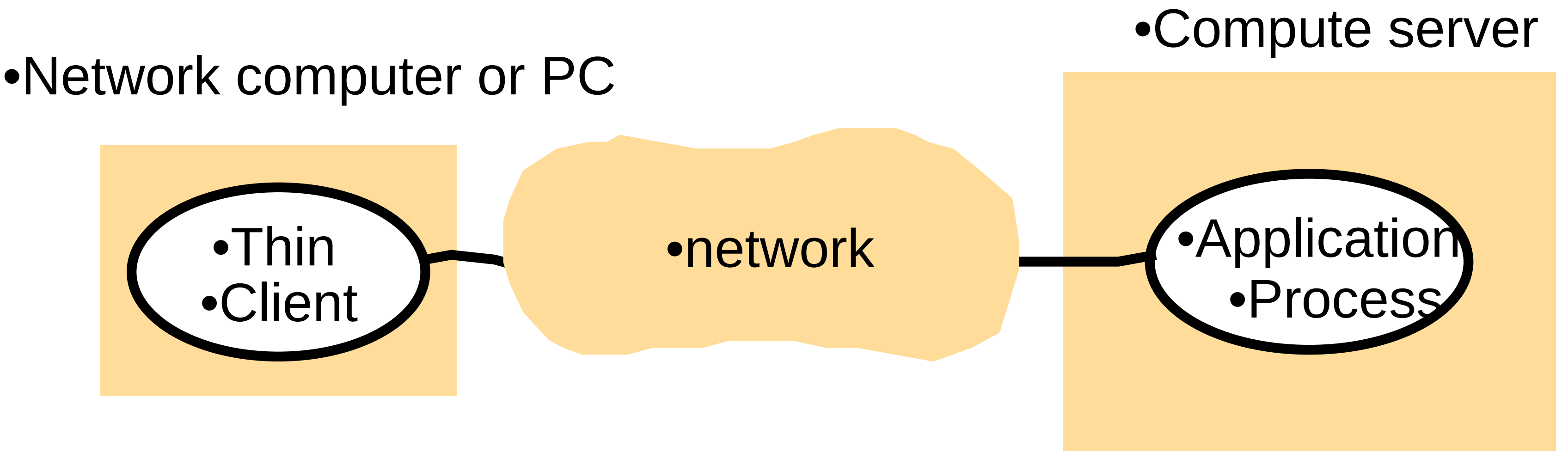
Client /Server



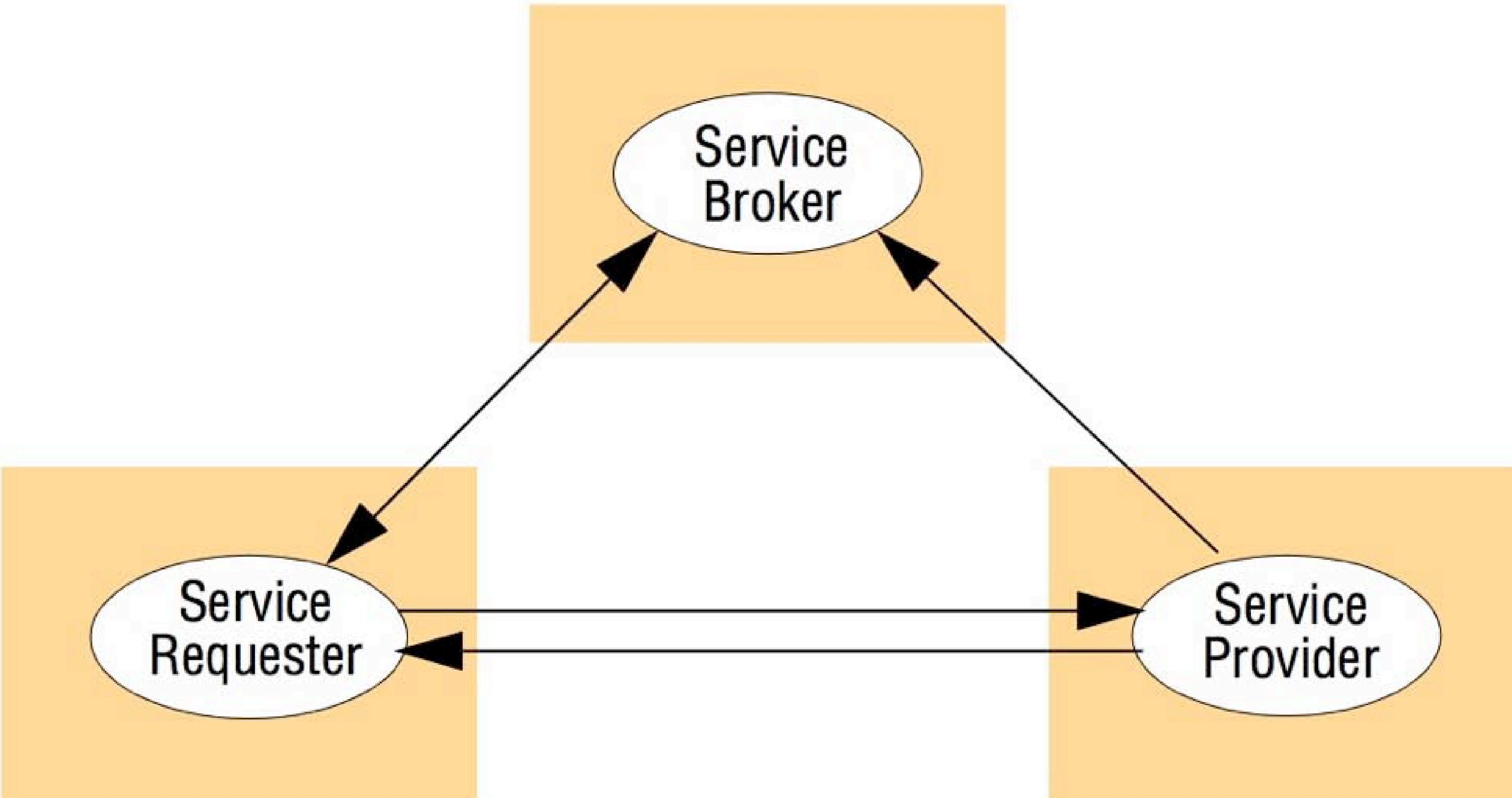
Client /Server



Client /Server

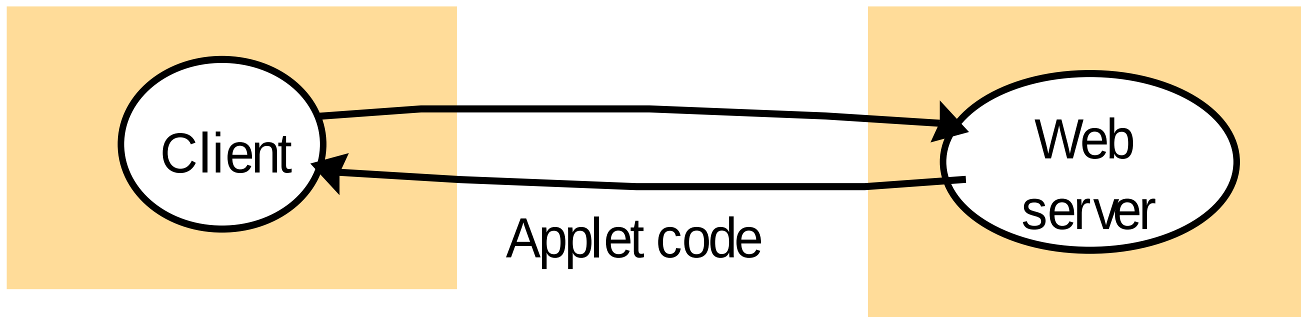


SOA Style



Mobile code

a) client request results in the downloading of applet code



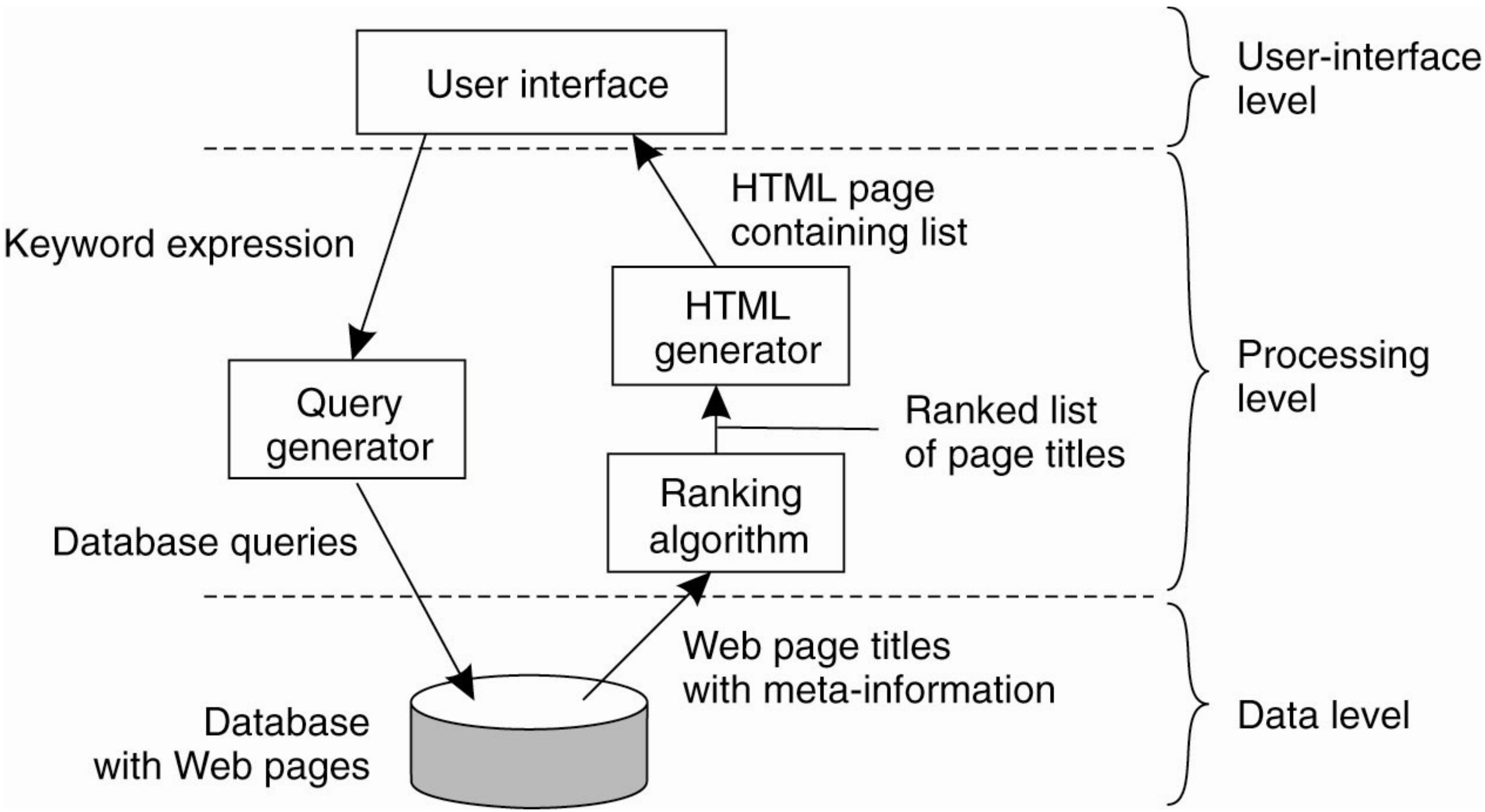
b) client interacts with the applet

Application Layering (1)

- Recall previously mentioned layers of architectural style
- The user-interface level
- The processing level
- The data level

Application Layering (2)

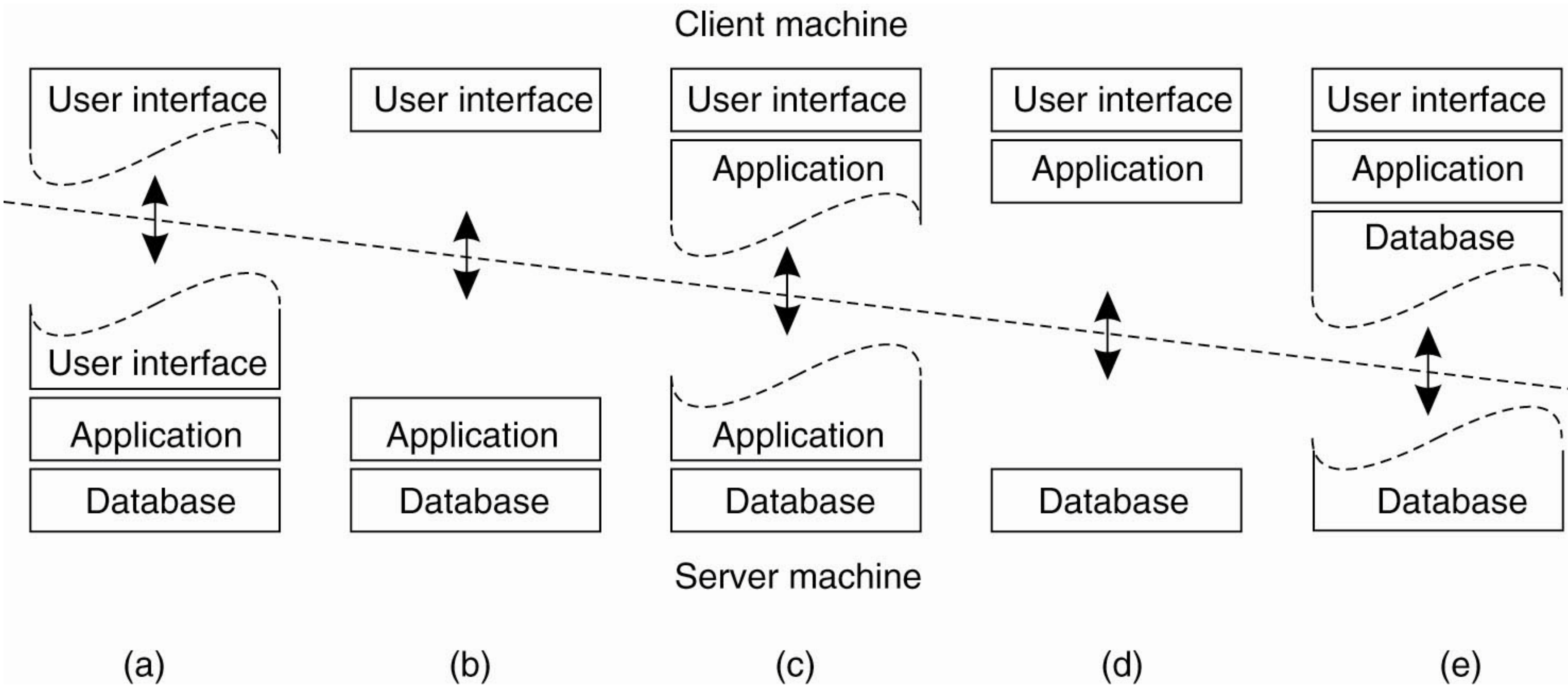
- The simplified organization of an Internet search engine into three different layers.



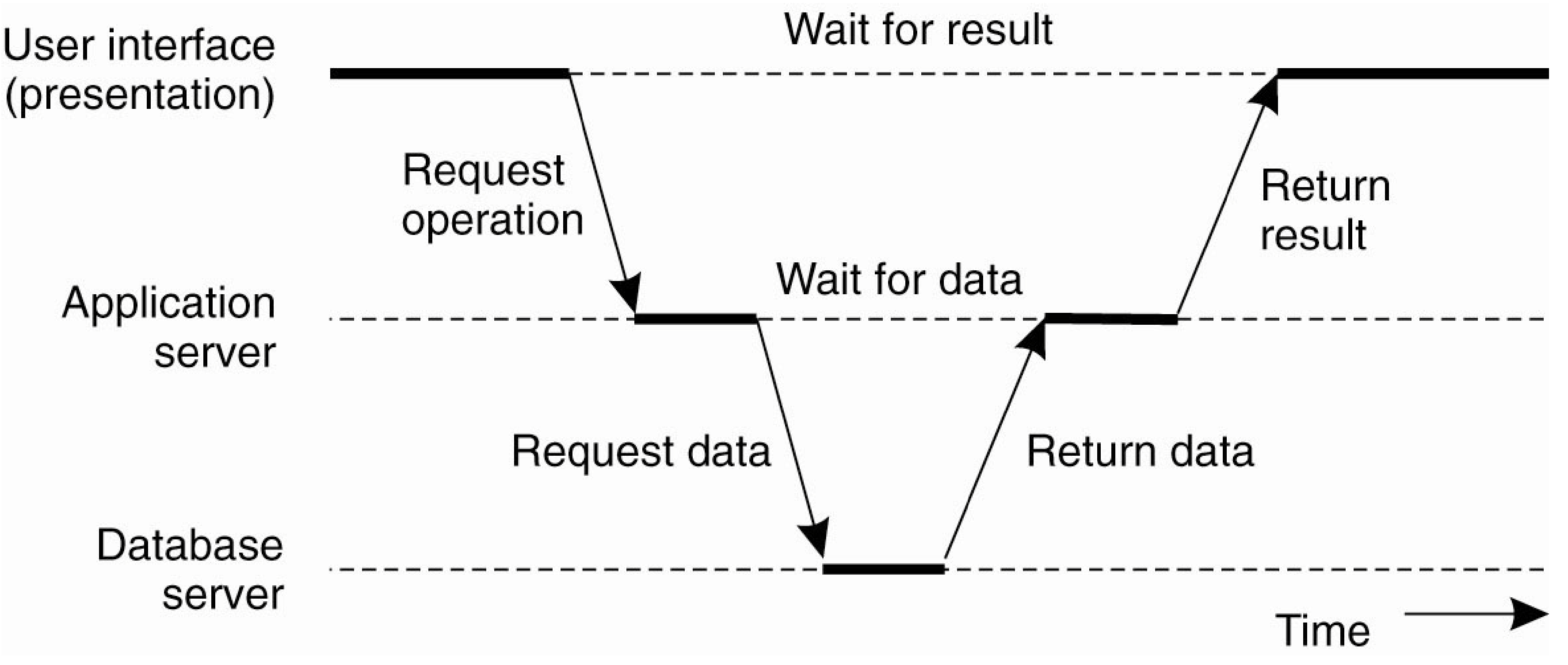
Multitiered Architectures (1)

- The simplest organization is to have only two types of machines:
- A client machine containing only the programs implementing (part of) the user-interface level
- A server machine containing the rest,
  - the programs implementing the processing and data level

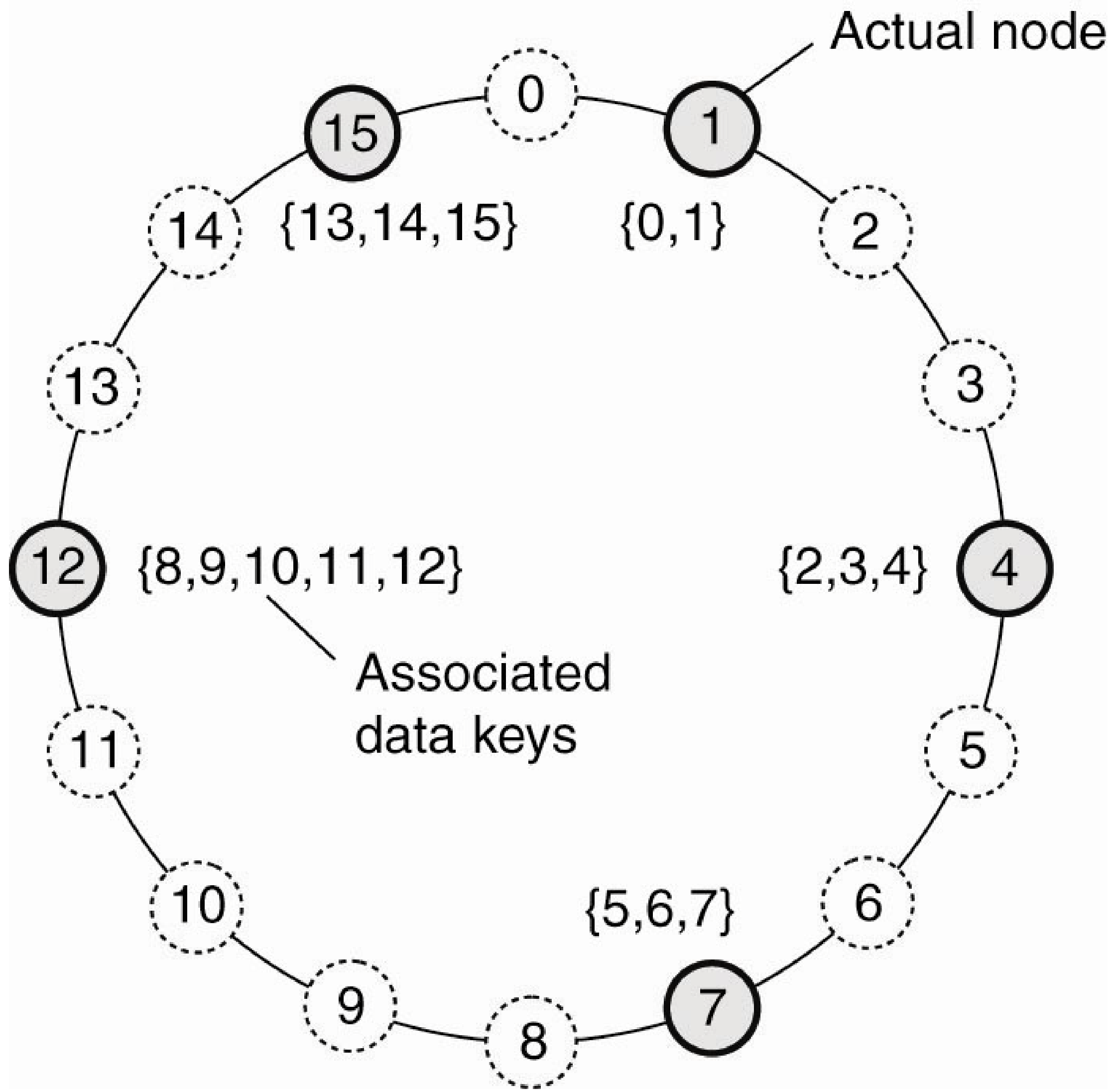
Multitiered Architectures (2)



Multitiered Architectures (3)



Structured Peer-to-Peer Architectures (1)



- The mapping of data items onto nodes in Chord.

Unstructured Peer-to-Peer Architectures (1)

Actions by active thread (periodically repeated):

```
select a peer P from the current partial view;
if PUSH_MODE {
    mybuffer = [(MyAddress, 0)];
    permute partial view;
    move H oldest entries to the end;
    append first c/2 entries to mybuffer;
    send mybuffer to P;
} else {
    send trigger to P;
}
if PULL_MODE {
    receive P's buffer;
}
construct a new partial view from the current one and P's buffer;
increment the age of every entry in the new partial view;
```

(a)

- The steps taken by the active thread.

Unstructured Peer-to-Peer Architectures (2)

Actions by passive thread:

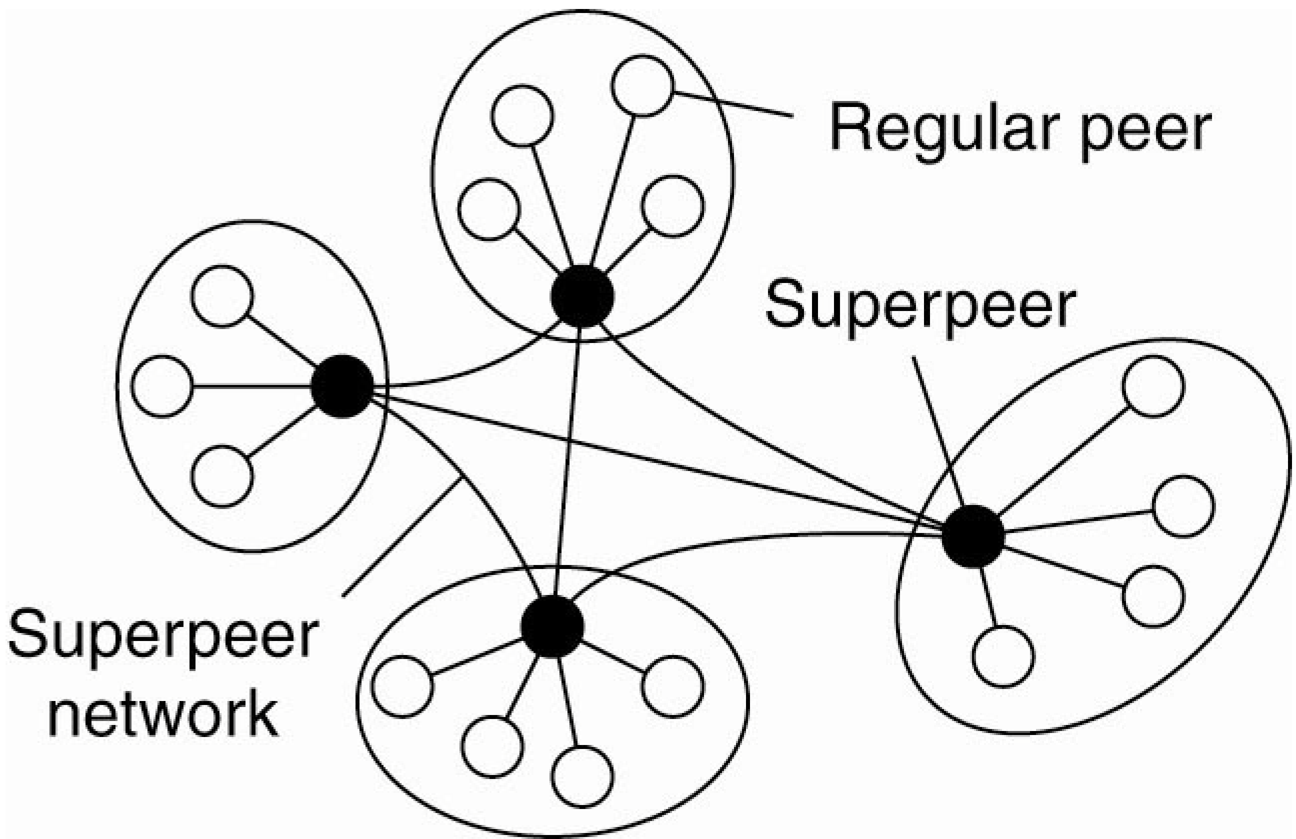
```
receive buffer from any process Q;
if PULL_MODE {
      mybuffer = [(MyAddress, 0)];
      permute partial view;
      move H oldest entries to the end;
      append first c/2 entries to mybuffer;
      send mybuffer to P;
}
construct a new partial view from the current one and P's buffer;
increment the age of every entry in the new partial view;
```

(b)

- The steps take by the passive thread

Superpeers

- A hierarchical organization of nodes into a superpeer network.



Edge-Server Systems

- Viewing the Internet as consisting of a collection of edge servers.